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--           >>      Spring Maven Life Cycle
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Validate → Compile → Test → Package → Integration-Test → Verify → Install →
Deploy

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--           >>      Build Life Cycle
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-- Suppose we building a house start with foundation like build the walls and
add the roof then paint and decorate.

-- In Maven, the build process is already fixed. This fixed build process is
called the Maven Lifecycle.

-- The lifecycle is divided into multiple small steps, and those steps are
called phases.

1. mvn test → runs validate + compile + test
2. mvn package → runs validate + compile + test + package
3. mvn install → runs everything till install

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-- There are three standard lifecycles in Maven :

1 - Default Lifecycle 🏠
Used for building + testing + packaging + installing + deploying

2 - Clean Lifecycle 🧹
Used for deleting old build files (target folder).
Example : mvn clean

3 - Site Lifecycle 📄
Used for generating project documentation.
Example : mvn site

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--           >>      Maven Default LifeCycle
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-- Maven Default Lifecycle has many phases, but in real projects we mainly focus
on 8 core phases because they represent the main build flow.

1) validate

Purpose : Checks if the project is correctly configured.
Example : Ensures pom.xml has required dependencies, plugins, and project info.

2) compile

Purpose : Compiles the main source code.
Example : Converts src/main/java → .class files.

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3) test

Purpose : Runs unit tests.

Example : Runs JUnit/TestNG tests from src/test/java.

4) package

Purpose : Creates the final build file.

Example : Generates .jar or .war inside target/.

5) integration-test

Purpose : Runs integration tests in a real-like environment.

Example : Tests DB + API + multiple modules working together.

6) verify

Purpose : Final validation before publishing.

Example : Ensures all tests passed and build is stable/valid.

7) install

Purpose : Stores the built artifact in local Maven repo.

Example : Saves .jar into ~/.m2/repository for reuse.

8) deploy

Purpose : Publishes the artifact to a remote repo.

Example : Uploads to Nexus/Artifactory for team + CI/CD usage.

Maven Key Features : If we run any phase, Maven automatic runs all phases before it.

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    >>    Maven Clean Life Cycle  (another name says that Starting Fresh)  
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-- The Clean lifecycle is used to remove previously generated build files from the project.

-- This ensures every new build starts fresh, without leftover or conflicting files.

-- Phases in the Clean Lifecycle

1 pre-clean

Purpose : Runs setup tasks before cleaning starts.

Example : Prepares everything and logs a message before the actual cleaning starts.

2 clean

Purpose : Deletes build output files, mainly the target/ folder.

Example : Removes compiled .class files and packaged .jar files.

3 post-clean

Purpose : Runs tasks after cleaning is completed.

Example : Logging cleanup completion or resetting environment variables.

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--    >>    Site Lifecycle : Generating Documentation
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--    Creates professional, organized documentation for your project.

--    Maven can generate reports like code coverage, dependency analysis, and test
results.

--    Helps stakeholders or developers see / work the project status clearly.

--    Phases in Site Lifecycle

1 pre-site

Purpose : Runs tasks needed before site generation.
Example : Prepares metadata for documentation.
(like - project names, version, description, report settings)

2 site

Purpose : Generates the project's website/documentation in target/site/.
Example : Produces reports on dependencies, code quality, and test results.

3 post-site

Purpose : Runs / Executes tasks after site generation.
Example : Logs or validates the generated site.
("Site generation completed or Verifies that the site was generated correctly or
not")

4 site-deploy

Purpose : Deploys the generated site to a remote server or repository.
(access team members or lead)
Example : Uploads documentation to a web server.

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Conclusion : Clean, Default, and Site Lifecycles in Harmony

--    Maven's clean, default, and site lifecycles work together to streamline
project management.

1 Clean : Start fresh by deleting old build files. (Clear the table)
2 Default : Build, test, and package the application. (Cook and serve food)
3 Site : Generate meaningful project documentation for stakeholders. (Make a
menu/report for guests)

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